

OKTYABR'SKOE, Russian Federation

WMO: 237340

Lat: 62.450N

Lon: 66.050E

Elev: 72

StdP: 100.46

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.4	-34.5	-40.2	0.1	-37.2	-37.3	0.1	-34.5	5.5	-16.8	4.8	-16.2	0.5	130	0.639

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.7	27.5	19.6	25.6	18.5	23.7	17.5	20.7	25.6	19.5	24.2	18.3	22.6	2.0	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	13.7	23.4	17.7	12.8	22.1	16.5	11.9	20.9	59.9	25.7	56.0	24.3	52.2	22.7	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.7	4.6	4.0	-39.3	30.4	4.3	1.1	-42.4	31.2	-44.9	31.8	-47.3	32.5	-50.4	33.2		
(5)			-39.7	22.2	4.3	1.0	-42.8	22.9	-45.4	23.5	-47.8	24.0	-50.9	24.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.4	-20.1	-17.4	-8.4	-1.1	6.7	14.1	17.1	13.1	7.1	-0.2	-11.5	-17.6		
	DBStd	14.23	8.26	7.70	6.22	5.70	5.60	5.24	4.36	4.02	4.22	4.86	8.01	9.11		
	HDD10.0	4704	933	767	569	335	138	20	2	14	108	316	646	855		
	HDD18.3	7281	1192	1001	828	584	362	146	77	171	338	574	896	1113		
	CDD10.0	531	0	0	0	1	35	143	222	109	21	1	0	0		
	CDD18.3	66	0	0	0	0	1	19	39	7	0	0	0	0		
	CDH23.3	468	0	0	0	0	16	142	269	41	0	0	0	0		
	CDH26.7	81	0	0	0	0	1	28	46	6	0	0	0	0		
	WSAvg	1.7	1.3	1.4	1.8	2.0	2.1	2.0	1.8	1.7	1.7	1.6	1.5	1.3		
(15) Precipitation	PrecAvg	591	34	24	27	34	49	63	76	83	63	59	45	37		
	PrecMax	850	56	66	69	75	101	154	170	178	131	132	94	65		
	PrecMin	408	10	3	5	8	8	9	8	25	8	17	7	11		
	PrecStd	100	11	13	16	17	22	32	43	33	30	22	19	16		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.8	0.4	5.6	15.6	25.6	29.8	27.0	21.1	13.0	3.5	0.1			
		MCWB	-1.3	-0.5	1.7	9.1	15.3	19.3	20.8	20.3	16.7	10.0	2.7	-0.4		
	2%	DB	-3.8	-2.2	3.0	11.0	22.1	27.0	28.0	24.1	18.1	10.3	1.8	-1.6		
		MCWB	-4.2	-3.1	0.2	5.9	14.0	18.6	20.4	18.4	14.3	8.3	1.1	-2.1		
	5%	DB	-6.1	-4.6	1.2	8.4	19.0	24.7	26.4	21.8	15.5	8.3	0.3	-3.7		
		MCWB	-6.5	-5.3	-0.6	4.5	12.3	17.2	19.4	17.2	12.3	6.8	-0.2	-4.1		
	10%	DB	-8.7	-6.8	-0.2	6.4	15.9	22.4	24.6	19.6	13.3	6.1	-1.1	-5.6		
		MCWB	-9.0	-7.4	-1.5	3.2	10.5	16.3	18.4	15.8	11.0	5.0	-1.6	-6.0		
	0.4%	WB	-1.0	-0.5	2.1	10.0	16.2	21.1	22.3	21.5	17.0	10.5	2.7	-0.3		
		MCDB	-0.9	0.2	4.9	14.8	23.0	26.4	27.4	26.1	20.4	12.3	3.0	0.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-4.1	-3.0	0.6	6.8	14.4	19.5	21.2	19.4	14.5	8.6	1.1	-2.0		
		MCDB	-3.8	-2.2	2.2	10.0	21.1	24.8	26.4	23.0	17.4	10.0	1.5	-1.6		
	5%	WB	-6.5	-5.2	-0.4	5.0	12.7	18.2	20.1	17.8	12.8	6.8	-0.2	-4.0		
		MCDB	-6.1	-4.6	1.0	7.9	17.9	23.6	25.2	20.7	14.8	8.1	0.3	-3.7		
	10%	WB	-9.0	-7.4	-1.6	3.5	11.0	16.8	19.1	16.4	11.3	5.1	-1.6	-5.9		
		MCDB	-8.7	-6.8	-0.2	6.1	15.9	21.7	23.7	19.0	13.2	6.0	-1.1	-5.6		
	5% DB	MDBR	6.4	6.9	8.1	8.6	9.4	9.0	8.7	7.6	6.5	5.0	5.7	6.6		
		MCDBR	6.3	6.6	8.3	10.5	12.9	11.6	10.7	10.3	9.1	7.3	4.7	6.8		
		MCWBR	6.4	6.3	6.5	6.9	6.9	5.8	5.3	5.6	5.7	5.4	4.7	6.7		
		MCWBR	6.6	6.7	7.8	9.4	11.9	11.0	9.8	9.4	8.5	7.2	5.1	6.5		
(40) Clear-Sky Solar Irradiance	taub	0.238	0.290	0.335	0.366	0.373	0.390	0.428	0.399	0.348	0.311	0.246	0.229			
	taud	1.957	2.033	2.016	2.092	2.332	2.371	2.248	2.309	2.423	2.316	2.023	1.908			
	Ebn at Noon	393	630	740	806	843	833	782	773	753	626	378	202			
	Edh at Noon	37	74	111	127	110	108	119	103	76	57	34	21			
(44) All-Sky Solar Radiation	RadAvg	0.28	1.12	2.64	4.24	4.96	5.31	5.19	3.63	2.12	0.94	0.32	0.13			
	RadStd	0.02	0.09	0.17	0.48	0.41	0.31	0.47	0.33	0.27	0.12	0.04	0.01			

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	+0.88	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page